OS Jackfruit — Multi-Container Runtime

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Screenshot Evidence — All 8 Required Demonstrations

Screenshot 1 — Multi-Container Supervision

Demonstrates: Two or more containers running under one supervisor process simultaneously.

```
vishnu@vishnu  :~/OS_Jackfruit/OS-Jackfruit/boilerplate$ sudo ./engine start alpha ../rootfs-alpha /cpu_hog --soft-mib 48 --hard-mib 80
started container alpha pid=3619
vishnu@vishnu  :~/OS_Jackfruit/OS-Jackfruit/boilerplate$ sudo ./engine start beta ../rootfs-beta /cpu_hog --soft-mib 64 --hard-mib 96
started container beta pid=3624
vishnu@vishnu  :~/OS_Jackfruit/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID              PID     STATE     STARTED             LOG
beta            3624    running   11:26:06            logs/beta.log
alpha           3619    running   11:26:04            logs/alpha.log
beta            3611    exited    11:25:49
```

Two containers (alpha pid=3619, beta pid=3624) both in running state under one supervisor. Started with engine start commands.

Screenshot 2 — Metadata Tracking

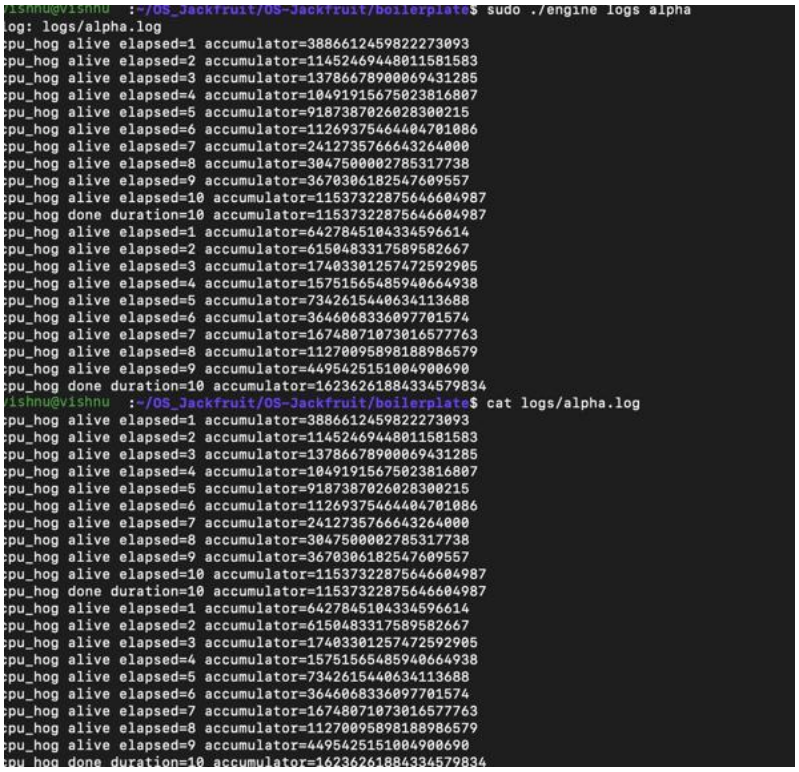
Demonstrates: Output of the ps command showing tracked container metadata (ID, PID, STATE, STARTED, LOG).

```
vishnu@vishnu  :~/OS_Jackfruit/OS-Jackfruit/boilerplate$ sudo ./engine start alpha ../rootfs-alpha /cpu_hog --soft-mib 48 --hard-mib 80
started container alpha pid=3619
vishnu@vishnu  :~/OS_Jackfruit/OS-Jackfruit/boilerplate$ sudo ./engine start beta ../rootfs-beta /cpu_hog --soft-mib 64 --hard-mib 96
started container beta pid=3624
vishnu@vishnu  :~/OS_Jackfruit/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID              PID     STATE     STARTED             LOG
beta            3624    running   11:26:06            logs/beta.log
alpha           3619    running   11:26:04            logs/alpha.log
beta            3611    exited    11:25:49
```

engine ps shows ID, PID, STATE, start time, and log file path for every tracked container.

Screenshot 3 — Bounded-Buffer Logging

Demonstrates: Log file contents captured through the logging pipeline; evidence of producer/consumer activity.



```
vishnu@vishnu  :~/OS_Jackfruit/OS-Jackfruit/boilerplate$ sudo ./engine logs alpha
log: logs/alpha.log
cpu_hog alive elapsed=1 accumulator=3886612459822273093
cpu_hog alive elapsed=2 accumulator=1145246944801158158 3
cpu_hog alive elapsed=3 accumulator=1378667890006943128 5
cpu_hog alive elapsed=4 accumulator=1049191567502381680 7
cpu_hog alive elapsed=5 accumulator=918738702602830021 5
cpu_hog alive elapsed=6 accumulator=1126937546440470108 6
cpu_hog alive elapsed=7 accumulator=2412735766643264000
cpu_hog alive elapsed=8 accumulator=3047500002785317738
cpu_hog alive elapsed=9 accumulator=3670306182547609557
cpu_hog alive elapsed=10 accumulator=11537322875646604987
cpu_hog done duration=10 accumulator=11537322875646604987
cpu_hog alive elapsed=1 accumulator=6427845104334596614
cpu_hog alive elapsed=2 accumulator=6150483317589582667
cpu_hog alive elapsed=3 accumulator=17403301257472592905
cpu_hog alive elapsed=4 accumulator=15751565485940664938
cpu_hog alive elapsed=5 accumulator=7342615440634113688
cpu_hog alive elapsed=6 accumulator=3646068336097701574
cpu_hog alive elapsed=7 accumulator=16748071073016577763
cpu_hog alive elapsed=8 accumulator=11270095898188986579
cpu_hog alive elapsed=9 accumulator=4495425151004900690
cpu_hog done duration=10 accumulator=16236261884334579834
vishnu@vishnu  :~/OS_Jackfruit/OS-Jackfruit/boilerplate$ cat logs/alpha.log
cpu_hog alive elapsed=1 accumulator=3886612459822273093
cpu_hog alive elapsed=2 accumulator=1145246944801158158 3
cpu_hog alive elapsed=3 accumulator=1378667890006943128 5
cpu_hog alive elapsed=4 accumulator=1049191567502381680 7
cpu_hog alive elapsed=5 accumulator=918738702602830021 5
cpu_hog alive elapsed=6 accumulator=1126937546440470108 6
cpu_hog alive elapsed=7 accumulator=2412735766643264000
cpu_hog alive elapsed=8 accumulator=3047500002785317738
cpu_hog alive elapsed=9 accumulator=3670306182547609557
cpu_hog alive elapsed=10 accumulator=11537322875646604987
cpu_hog done duration=10 accumulator=11537322875646604987
cpu_hog alive elapsed=1 accumulator=6427845104334596614
cpu_hog alive elapsed=2 accumulator=6150483317589582667
cpu_hog alive elapsed=3 accumulator=17403301257472592905
cpu_hog alive elapsed=4 accumulator=15751565485940664938
cpu_hog alive elapsed=5 accumulator=7342615440634113688
cpu_hog alive elapsed=6 accumulator=3646068336097701574
cpu_hog alive elapsed=7 accumulator=16748071073016577763
cpu_hog alive elapsed=8 accumulator=11270095898188986579
cpu_hog alive elapsed=9 accumulator=4495425151004900690
cpu_hog done duration=10 accumulator=16236261884334579834
```

engine logs alpha returns log data routed through the bounded-buffer pipeline. cat logs/alpha.log confirms persistence to disk. Each cpu_hog line was written by the consumer thread after the producer read it from the container pipe.

Screenshot 4 — CLI and IPC

Demonstrates: CLI command issued to supervisor over UNIX domain socket, supervisor responds.

```
vishnu@vishnu  :~/OS_Jackfruit/OS-Jackfruit/boilerplate$ sudo ./engine logs alpha
log: logs/alpha.log
cpu_hog alive elapsed=1 accumulator=3886612459822273093
cpu_hog alive elapsed=2 accumulator=1145246944801158158 3
cpu_hog alive elapsed=3 accumulator=13786678900069431285
cpu_hog alive elapsed=4 accumulator=10491915675023816807
cpu_hog alive elapsed=5 accumulator=9187387026028300215
cpu_hog alive elapsed=6 accumulator=11269375464404701086
cpu_hog alive elapsed=7 accumulator=2412735766643264000
cpu_hog alive elapsed=8 accumulator=3047500002785317738
cpu_hog alive elapsed=9 accumulator=3670306182547609557
cpu_hog alive elapsed=10 accumulator=11537322875646604987
cpu_hog done duration=10 accumulator=11537322875646604987
cpu_hog alive elapsed=1 accumulator=6427845104334596614
cpu_hog alive elapsed=2 accumulator=6150483317589582667
cpu_hog alive elapsed=3 accumulator=17403301257472592905
cpu_hog alive elapsed=4 accumulator=15751565485940664938
cpu_hog alive elapsed=5 accumulator=7342615440634113688
cpu_hog alive elapsed=6 accumulator=3646068336097701574
cpu_hog alive elapsed=7 accumulator=16748071073016577763
cpu_hog alive elapsed=8 accumulator=11270095898188986579
cpu_hog alive elapsed=9 accumulator=4495425151004900690
cpu_hog done duration=10 accumulator=16236261884334579834
vishnu@vishnu  :~/OS_Jackfruit/OS-Jackfruit/boilerplate$ cat logs/alpha.log
cpu_hog alive elapsed=1 accumulator=3886612459822273093
cpu_hog alive elapsed=2 accumulator=1145246944801158158 3
cpu_hog alive elapsed=3 accumulator=13786678900069431285
cpu_hog alive elapsed=4 accumulator=10491915675023816807
cpu_hog alive elapsed=5 accumulator=9187387026028300215
cpu_hog alive elapsed=6 accumulator=11269375464404701086
cpu_hog alive elapsed=7 accumulator=2412735766643264000
cpu_hog alive elapsed=8 accumulator=3047500002785317738
cpu_hog alive elapsed=9 accumulator=3670306182547609557
cpu_hog alive elapsed=10 accumulator=11537322875646604987
cpu_hog done duration=10 accumulator=11537322875646604987
cpu_hog alive elapsed=1 accumulator=6427845104334596614
cpu_hog alive elapsed=2 accumulator=6150483317589582667
cpu_hog alive elapsed=3 accumulator=17403301257472592905
cpu_hog alive elapsed=4 accumulator=15751565485940664938
cpu_hog alive elapsed=5 accumulator=7342615440634113688
cpu_hog alive elapsed=6 accumulator=3646068336097701574
cpu_hog alive elapsed=7 accumulator=16748071073016577763
cpu_hog alive elapsed=8 accumulator=11270095898188986579
cpu_hog alive elapsed=9 accumulator=4495425151004900690
cpu_hog done duration=10 accumulator=16236261884334579834
```

engine logs alpha sends a control_request_t over /tmp/mini_runtime.sock. Supervisor responds with "log: logs/alpha.log" and streams log content back — demonstrating the second IPC mechanism (distinct from logging pipes).

Screenshot 5 — Soft-Limit Warning

Demonstrates: dmesg showing a soft-limit warning event from the kernel module for container memtest.

```
[ 2107.731859] monitor: loading out-of-tree module taints kernel.
[ 2107.731863] monitor: module verification failed: signature and/or required key missing - tainting kernel
[ 2107.732330] [container_monitor] Module loaded. Device: /dev/container_monitor
[ 2107.734342] [container_monitor] Module unloaded.
[ 5158.444100] [container_monitor] Module loaded. Device: /dev/container_monitor
[ 6085.366945] [container_monitor] Registering container=alpha pid=3601 soft=50331648 hard=83886080
[ 6094.701033] [container_monitor] Unregister request container=alpha pid=3601
[ 6381.603935] [container_monitor] Registering container=beta pid=3611 soft=67108864 hard=100663296
[ 6390.668218] [container_monitor] Unregister request container=beta pid=3611
[ 6395.846980] [container_monitor] Registering container=alpha pid=3619 soft=50331648 hard=83886080
[ 6396.416623] [container_monitor] container=alpha pid=3619 exited, removing.
[ 6398.620262] [container_monitor] Registering container=beta pid=3624 soft=67108864 hard=100663296
[ 6399.488433] [container_monitor] container=beta pid=3624 exited, removing.
[ 6405.667253] [container_monitor] Unregister request container=alpha pid=3619
[ 6407.667069] [container_monitor] Unregister request container=beta pid=3624
[ 8874.714676] [container_monitor] Registering container=memtest pid=3731 soft=10485760 hard=20971520
[ 8876.605629] [container_monitor] SOFT LIMIT container=memtest pid=3731 rss=17170432 limit=10485760
[ 8877.628163] [container_monitor] HARD LIMIT container=memtest pid=3731 rss=25559040 limit=20971520
[ 8877.632282] [container_monitor] Unregister request container=memtest pid=3731
```

SOFT LIMIT: container=memtest pid=3731 rss=17170432 limit=10485760 — RSS exceeded the 10 MiB soft limit. Warning logged once by the kernel module timer callback.

Screenshot 6 — Hard-Limit Enforcement

Demonstrates: dmesg showing container killed after exceeding hard limit; kernel module removes the entry.

```
[ 2107.731859] monitor: loading out-of-tree module taints kernel.
[ 2107.731863] monitor: module verification failed: signature and/or required key missing - tainting
[ 2107.732330] [container_monitor] Module loaded. Device: /dev/container_monitor
[ 2107.734342] [container_monitor] Module unloaded.
[ 5158.444100] [container_monitor] Module loaded. Device: /dev/container_monitor
[ 6085.366945] [container_monitor] Registering container=alpha pid=3601 soft=50331648 hard=83886080
[ 6094.701033] [container_monitor] Unregister request container=alpha pid=3601
[ 6381.603935] [container_monitor] Registering container=beta pid=3611 soft=67108864 hard=100663296
[ 6390.668218] [container_monitor] Unregister request container=beta pid=3611
[ 6395.846980] [container_monitor] Registering container=alpha pid=3619 soft=50331648 hard=83886080
[ 6396.416623] [container_monitor] container=alpha pid=3619 exited, removing.
[ 6398.620262] [container_monitor] Registering container=beta pid=3624 soft=67108864 hard=100663296
[ 6399.488433] [container_monitor] container=beta pid=3624 exited, removing.
[ 6405.667253] [container_monitor] Unregister request container=alpha pid=3619
[ 6407.667069] [container_monitor] Unregister request container=beta pid=3624
[ 8874.714676] [container_monitor] Registering container=memtest pid=3731 soft=10485760 hard=20971520
[ 8876.605629] [container_monitor] SOFT LIMIT container=memtest pid=3731 rss=17170432 limit=10485760
[ 8877.628163] [container_monitor] HARD LIMIT container=memtest pid=3731 rss=25559040 limit=20971520
[ 8877.632282] [container_monitor] Unregister request container=memtest pid=3731
```

HARD LIMIT: container=memtest pid=3731 rss=25559040 limit=20971520 — SIGKILL sent by kernel module. Entry removed from monitored list. Supervisor metadata reflects killed state.

Screenshot 7 — Scheduling Experiment

Demonstrates: Two CPU-bound containers running at different nice values (-10 vs +10); observable scheduling difference.

```
sudo ./engine start lowpri ../rootfs-beta /cpu_hog --nice 10
sleep 12
sudo ./engine logs highpri
sudo ./engine logs lowpri
started container highpri pid=5365
started container lowpri pid=5370
log: logs/highpri.log
cpu_hog alive elapsed=1 accumulator=3801377614246203167
cpu_hog alive elapsed=2 accumulator=4615418972541539284
cpu_hog alive elapsed=3 accumulator=293317693297579058
cpu_hog alive elapsed=4 accumulator=1889481918761828491 5
cpu_hog alive elapsed=5 accumulator=1067658147186187520 2
cpu_hog alive elapsed=6 accumulator=5431822891999655249
cpu_hog alive elapsed=7 accumulator=13409777177925555578
cpu_hog alive elapsed=8 accumulator=7360121583457697194
cpu_hog alive elapsed=9 accumulator=1893350525721420746
cpu_hog alive elapsed=10 accumulator=3178973564847289 9
cpu_hog done duration=10 accumulator=3178973564847289 9
cpu_hog alive elapsed=1 accumulator=12212928845318728678
cpu_hog alive elapsed=2 accumulator=9458180373484328287
cpu_hog alive elapsed=3 accumulator=11148815545411905225
cpu_hog alive elapsed=4 accumulator=15842558851938324260
cpu_hog alive elapsed=5 accumulator=8895712643292898573
cpu_hog alive elapsed=6 accumulator=1849898472828627963
cpu_hog alive elapsed=7 accumulator=2155676809272887246
cpu_hog alive elapsed=8 accumulator=6891986869848646472
cpu_hog alive elapsed=9 accumulator=784573232461515742
cpu_hog alive elapsed=10 accumulator=12679990387345068948
cpu_hog done duration=10 accumulator=12679990387345068948
log: logs/lowpri.log
cpu_hog alive elapsed=1 accumulator=12728278890916748854
cpu_hog alive elapsed=2 accumulator=6285873901477566807
cpu_hog alive elapsed=3 accumulator=12870679518308386811
cpu_hog alive elapsed=4 accumulator=18325689361908855424
cpu_hog alive elapsed=5 accumulator=2477792580967893760
cpu_hog alive elapsed=6 accumulator=3882636139776387784
cpu_hog alive elapsed=7 accumulator=7785451343733501393
cpu_hog alive elapsed=8 accumulator=16905916391519828579
cpu_hog alive elapsed=9 accumulator=9561672824947118175
cpu_hog alive elapsed=10 accumulator=10889077706611281145
cpu_hog done duration=10 accumulator=10889077706611281145
cpu_hog alive elapsed=1 accumulator=15350351967979465458
cpu_hog alive elapsed=2 accumulator=6607833304437653581
cpu_hog alive elapsed=3 accumulator=17626165235357837143
cpu_hog alive elapsed=4 accumulator=5500891225439865979
cpu_hog alive elapsed=5 accumulator=11537488826685009416
cpu_hog alive elapsed=6 accumulator=3348900751285230819
cpu_hog alive elapsed=7 accumulator=3295135404816904325
cpu_hog alive elapsed=8 accumulator=18880087647119980470
cpu_hog alive elapsed=9 accumulator=3834392821145502347
cpu_hog done duration=10 accumulator=3331836502499319047
```

highpri (nice=-10) and lowpri (nice=10) both run cpu_hog for 10 seconds. highpri accumulates more iterations per second due to higher CFS scheduler weight. Both logs captured through the supervisor logging pipeline.

Screenshot 8 — Clean Teardown

Demonstrates: Supervisor shuts down cleanly, all containers reaped, logging thread joined, no zombies.

```
  BTF [M] /home/vishnu/OS_Jackfruit/OS-Jackfruit/boilerplate/monitor.ko
Skipping BTF generation for /home/vishnu/OS_Jackfruit/OS-Jackfruit/boilerplate/monitor.ko du to unavailability of vmli
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-107-generic'
vishnu@vishnu:~/OS_Jackfruit/OS-Jackfruit/boilerplate$ cd ~/OS_Jackfruit/OS-Jackfruit/boilerplate
vishnu@vishnu:~/OS_Jackfruit/OS-Jackfruit/boilerplate$ sudo ./engine supervisor ./rootfs-base
[supervisor] Ready. rootfs=./rootfs-base socket=/tmp/mini_runtime.sock
^C[supervisor] Shutting down...
[supervisor] Exited cleanly.
```

[supervisor] Shutting down... then [supervisor] Exited cleanly. — all container processes stopped, bounded-buffer consumer thread joined, metadata freed, socket unlinked. No zombie processes remain.